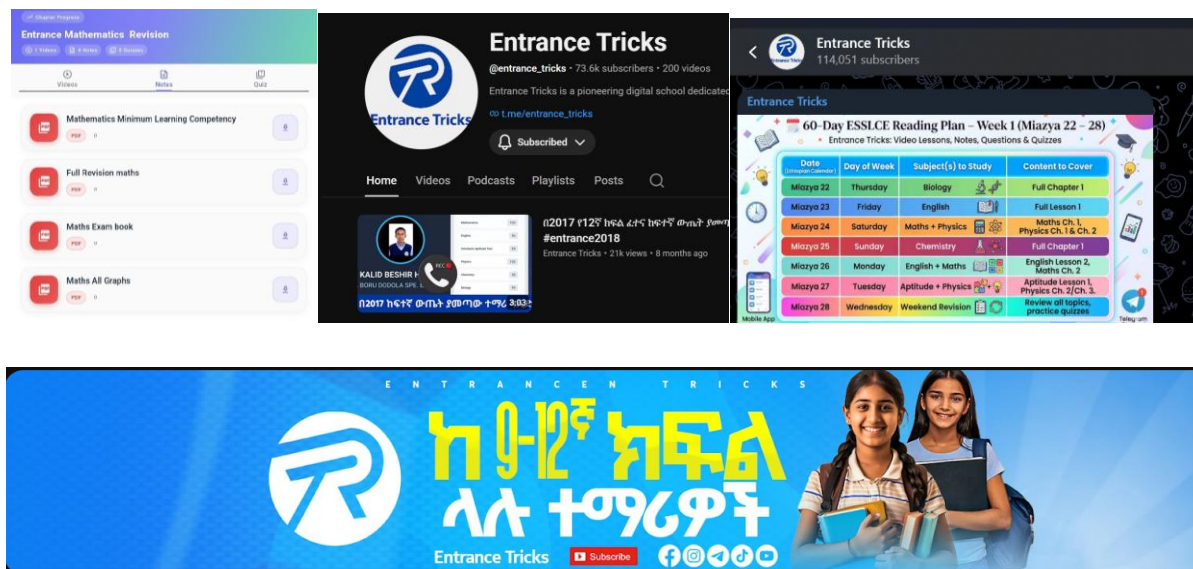


2018 Biology First Round ESSLCE Online Exam



1. A city Municipality following the ban of non-degradable plastic bags, proposed to use genetically engineered bacterium to degrade plastic wastes for making bags. However, an authority claims the use of bioengineered bacteria due to the fear of its negative consequences on the environment. Based on this scenario, which decision is valid?

- A. accepted with stringent protocols to reduce its effect.
- B. rejected since the city can use already existing materials for packing.
- C. accepted whatever the environmental consequences it caused.
- D. rejected since the municipality can use the native species.

2. This item is based on the following mechanism of hearing in human ear.

- I. Nerve impulses produced by hair-like sensory cells are transmitted to, and interpreted by the brain.
- II. Ear ossicles amplify vibrations and cause vibration of the ear fluid.
- III. Sound wave from the air is collected by pinna and directs to the ear canal.
- IV. Vibrations are magnified in eardrum and transmitted to oval window.

Then which of the following order summarizes correct path of hearing in human ear?

- A. III, I, II, IV
- B. III, II, IV, I
- C. III, IV, II, I
- D. III, IV, I, II

3. Which one of the following is the industrial importance of insects?

- A. Serving as pollinating agents for plants
- B. Producing silk fibers for textiles
- C. Controlling pests that attack crops
- D. Serving as the source of pain-relieving drugs

4. Two processes were taking place inside a eukaryotic cell. In process A, a ribosome made a specific protein by reading the instructions from mRNA. In process B, an mRNA was synthesized inside the nucleus. Based on this scenario, which of the following processes correctly

represents A and B respectively?

- A. Translation and Replication
- B. Transcription and Translation
- C. Replication and Translation
- D. Translation and Transcription

5. Which one of the following statements is correct about regulation of enzyme through covalent modification? It happens through the

- A. permanent binding of effectors to the active site of an enzyme.
- B. addition or elimination of some molecules attaching of enzyme protein.
- C. reversible binding of effectors to the allosteric site of an enzyme.
- D. inhibition of the enzyme by the end product of a pathway.

6. This item is based on the following table that describes about the role of digestive enzymes and their optimum pH.

No.	Enzymes	The role of the enzyme	Optimum pH
1.	Pepsin	Hydrolyzes proteins to amino acids	1
2.	Amylase	Converts starch to maltose	4 - 6
3.	Nuclease	Converts nucleotides to nucleic acids	1
4.	Lactase	Hydrolyzes lactose to glucose and galactose	7 - 8

Which alternative contains the correct role of the enzyme and its optimum pH?

- A. 2 and 4
- B. 1 and 4
- C. 1, 2 and 4
- D. 1, 3 and 4

7. A farmer crossed a heterozygous insect-resistant maize with a non-insect resistant maize variety, where insect-resistance gene is dominant over non-resistance one. What is the probability of the offspring resulted from this cross?

- A. 75% resistant and 25% are non-resistant to insect infestation.
- B. 50% resistant and 50% are non-resistant to insect infestation.
- C. 25% resistant and 75% are non-resistant to insect infestation.
- D. 100% of the offspring are resistant to insect infestation.

8. Which statement does NOT show the role of biological discoveries for the society?

- A. Detection of cancer cells through advanced instruments
- B. Collecting and analyzing evidences from crime scene
- C. Studying of microorganisms that cause disease to humans
- D. Resolving conflict that happens as a result of over population

9. The function of ovaries in female reproductive system differs from testes in the male reproductive system in which ovaries produce

- A. estrogen that initiates pregnancy, whereas testes produce LH and sperm.
B. progesterone that initiates ovulation, whereas testes produce FSH and sperm.
C. estrogen that promotes the maturation of the ovum, whereas testes produce testosterone and sperm.
D. androgen that develops a mature ovum, whereas testes produce estrogen and sperm.
- 11.** A following biology guide prepared a thin slice of onion epidermal cells and placed it in a hypertonic solution for 15 minutes. What would be the result of the experiment? They
- A. become crenated due to damage to the cell walls by the water pressure.
B. burst as the vacuoles presses against the cytoplasm and cell membrane.
C. get plasmolyzed due to excessive loss of water from the vacuole.
D. become turgid due to net inflow of water from the external solution.
- 12.** Which of the following is correct regarding photorespiration? It is a process of
- A. balancing O_2 and CO_2 concentration in the cell.
B. oxidizing organic compounds to release CO_2 .
C. respiration common in plants during the daytime.
D. reduction of organic compounds to generate ATP.
- 13.** Which alternative contains the behavioral method of thermoregulation with its example?
- A. Honey bees - stay in hive to survive cold seasons
B. Cats - use involuntary muscle contraction to generate heat
C. Foxes - narrow blood vessels to prevent heat loss
D. Rodents - sweat to maintain body temperature
- 14.** Why hillside terracing is considered as a mechanical soil conservation measure? Because it
- A. is constructed to intercept runoff water that drain from the slopy area.
B. breaks largely the speed of wind to avoid wind erosion.
C. is a reservoir of rainwater to use the water for irrigation purpose.
D. is constructed to divert the rainwater to other area with higher slope.
- 15.** The amount of energy required to raise the temperature of one gram of water by one degree Celsius is
- A. cohesion force.
B. surface tension.
C. latent heat of vaporization.
D. specific heat capacity.
- 16.** This item is based on the following table about the classification of organic molecules.

No.	Organic molecules	Elemental composition	Monomers
1.	Sucrose	Carbon, hydrogen and oxygen	Glucose and fructose
2.	Starch	Carbon, hydrogen and oxygen	Glucose and galactose
3.	Proteins	Carbon, hydrogen and oxygen	Amino acids

No.	Organic molecules	Elemental composition	Monomers
4.	Lipids	Carbon, hydrogen and oxygen	Fatty acid and glycerol

Which numbers in the above table correspond to the correct elemental composition and monomers of the organic molecules?

- A. 1, 2 and 4
- B. 1, 3 and 4
- C. 1 and 3
- D. 1 and 4

17. What are ethical issues in biology? They are issues linked to

- A. use of microscope to observe human cheek cells.
- B. study the interaction between organisms.
- C. use of scientific discoveries for human wellbeing.
- D. study the role of evolution in speciation.

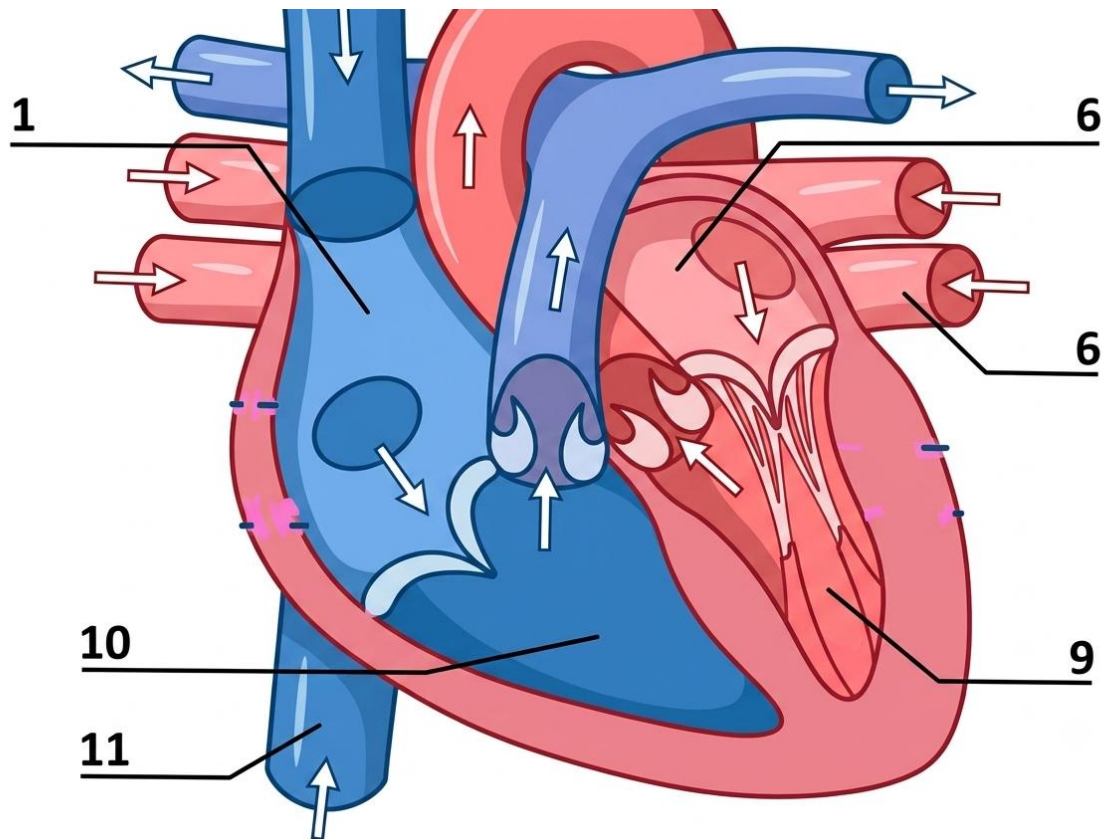
18. Which of the following is a symptom of gonorrhea infection?

- A. Painful urination and abnormal discharge from the penis
- B. Necrotizing genital ulcer accompanied by lymphadenopathy
- C. Sores around the anus that leads to cancer through time
- D. Rash on hands and soles of feet with late neurological complication

19. All the synthesis and breakdown chemical reactions together occurred in our cells is

- A. catabolism.
- B. anabolism.
- C. respiration.
- D. metabolism.

20. Based on the following diagram of the human heart, identify the flow and anatomical parts accordingly.



- A. Left atrium receives deoxygenated blood
- B. Right ventricle pumps oxygenated blood
- C. Aorta distributes oxygenated blood to the body
- D. Pulmonary vein carries deoxygenated blood

21. A biologist observed the sister chromatids that lined up end-to-end at the equator of the cell in a sample taken from Test tube 1, whereas in another cell sample (Test tube 2) she observed the sister chromatids are pulled to opposite poles of the equator. Based on this information, which stages of meiosis are observed in test tube 1 and 2?

- A. Prophase I and Metaphase I
- B. Metaphase II and Anaphase II
- C. Anaphase II and Telophase II
- D. Prophase I and Telophase I

22. Suppose the fibers in the connective tissue of the skin and blood vessels do not form properly and causes bleeding gums and poor healing of wounds. This problem is related with deficiency of

- A. Vitamin D.
- B. Vitamin A.
- C. Vitamin B.
- D. Vitamin C.

23. What would happen to a plant placed in areas with unilateral light?

- A. The light exposed side would produce more root hairs.
- B. The light exposed parts would produce more plant hormones.

- C. The plant would bend towards the light source.
D. The plant would immediately stop growth.
- 24.** In a grassland, the gazelle population initially increase in fast pace, then the growth rate reduces gradually and finally levels-off when the population reaches its maximum. This population growth model shows
A. exponential because the gazelle experiences maximum per capita rate of increase.
B. logistic because the size of gazelle population reaches the carrying capacity.
C. logistic because the per capita rate increase decreases as the population decreases.
D. exponential since the growth rate of the gazelle reach maximum at the carrying capacity.
- 25.** Grade 9 students identified an organism using dichotomous keys. The organism was poikilothermic, invertebrate and had six jointed legs. To which phylum would the students group this organism?
A. Arthropoda
B. Mollusca
C. Porifera
D. Cnidaria
- 26.** Which one of the following statements is correct about innate immunity? It
A. produces antibody that neutralizes antigens in the blood.
B. responds better after the second encounter of a pathogens.
C. has memory to recognize pathogens.
D. is a non-specific host defense system.
- 27.** Which of the following is NOT a pollinating agent?
A. Birds
B. Insects
C. Wind
D. Microbes
- 28.** Which of the following is the major causes of evolution that results from a sudden, and random change of DNA sequence?
A. Artificial selection
B. Induced mutation
C. Environmental stability
D. Acquired characteristic
- 29.** Gardener brings an ornamental plant from other country and plant it in urban parks. This species through time moved into conservation area and caused a loss for many other plant species. This is a loss of biodiversity due to
A. invasive species.
B. climate change.
C. deforestation.
D. overexploitation.
- 30.** Which one of the following breathing structures and its function is correctly matched?
A. Trachea - prevents the collapse of respiratory structures
B. Bronchioles - deliver air from the nasal cavity to the bronchi
C. Bronchi - exchange gases between capillaries in the lungs
D. Larynx - prevents choking by covering the opening in the trachea

31. Suppose a person's height and body mass are 1.81 m and 68 kg, respectively. Which one of the following categories best describes the person? She is

- A. overweight.
- B. normal.
- C. obese.
- D. underweight.

32. Which one of the following statements is correct about complete metamorphosis? The

- A. larvae develop directly into adult
- B. larvae develop into nymph
- C. eggs hatch into larvae after few days
- D. nymph develops into adult

33. Plant cells differ from animal cells in that, they have

- A. larger ribosomes for protein synthesis.
- B. large central vacuoles in their cells.
- C. cell membrane with double layers.
- D. a pair of centrioles for cell division.

34. This item is based on the following table about food chain.

No.	Characteristics	Trophic level	Feeding level	Example
1.	Make their own food	First	Producers	Grass
2.	Feeds on herbivores	Third	Tertiary consumer	Frog
3.	Feeds on another carnivore	Fourth	Tertiary consumer	Snake
4.	Feeds on green plants	Second	Primary consumer	Grasshopper

Which numbers in the above table contain the correct description of a food chain with its examples?

- A. 1, 2 and 3
- B. 1 and 2
- C. 2 and 4
- D. 1, 3 and 4

35. Which one of the following bacteria are correctly grouped into their category based on their shape?

- A. Bacilli - rod shaped bacteria
- B. Cholera - spiral shaped bacteria
- C. Spirochaetes - spherical bacteria
- D. Cocci - corkscrew shaped bacteria

36. Which one of the following nutrients is correctly paired with its deficiency diseases?

- A. Phosphorous - kwashiorkor
- B. Protein - marasmus

C. Calcium - rickets

D. Iron - bleeding gums

37. In the presence of oxygen, the cell theoretically yields approximately 36-38 ATP per glucose. Justify why the majority of this energy is harvested by electron transport system (ETS) rather than by glycolysis or the Krebs cycle? Because ETS

A. is the sole stage where glucose is directly converted into ATP through substrate level phosphorylation.

B. uses the stored potential energy of NADH and FADH_2 to create proton gradient that drives ATP synthase.

C. occurs in the cytosol where there is more space to produce ATP abundantly.

D. utilizes oxygen that plugs the leak in the membrane, leading to produce more metabolic energy.

38. Which of the following is correct about gene therapy? It is a genetic technique that helps to

A. replace the whole genome of a person with genetic disorder.

B. enhance transfer of genetic disorders from parents to their offsprings.

C. identify disorders that passes from parents to offsprings.

D. treat genetic disorders by transferring sections of DNA to patient's cells.

39. Which structure and function of the types of teeth is correctly matched?

A. Molars - broader teeth specialized for grinding foods

B. Premolars - flat teeth specialized for tearing foods

C. Incisors - flattened teeth specialized for crushing foods

D. Canine - sharp teeth specialized for cutting foods

40. Among the following features, which one describes kingdom fungi? They

A. are multicellular and have chemoautotrophic mode of nutrition.

B. ingest and digest their food by enzymes in their body.

C. undergo extracellular digestion to obtain their nutrition.

D. are prokaryotic and cosmopolitan in their distribution.

41. Which of the following is the correct definition of mutation? It is a

A. spontaneous variation in the spatial distribution of organisms.

B. random change in genetic information of organisms.

C. gradual change in the phenotype of the organisms.

D. gradual variation in the spatial distribution of organisms.

42. Suppose a biology teacher take a sample of bacteria from a food and dissolve in a distilled water. If she wants to isolate the bacterium, what is the next direct procedure to accomplish her task?

A. Conduct gram staining to identify their feature

B. Count a colony and harvest target product

C. Perform serial dilution and grow in media

D. Identify bacterial morphology and grow in media

43. In which of the following options, a hormone is correctly paired with its function?

A. Norepinephrine - facilitate the secretion of glucagon

B. Oxytocin - helps during delivery of the baby

C. Parathyroid - control iodine levels in blood stream

D. Aldosterone - increase heart rate and force of contraction

44. This item is based on the following events of maintaining water balance in the body.

- I. Second coiled tubules of kidney become permeable to reabsorb more water from the kidney
- II. High ADH is released into the blood
- III. Produce large volume of diluted urine due to less reabsorption of water by kidney
- IV. Very few ADH released to the blood

Based on the above list, which of the following numbers shows the mechanism of water balance in our body correctly? Numbers

- A. I and IV are for water conserving, II and III are for water removing.
- B. III and IV are for water conservation, I and II are for water removing.
- C. I and IV are for water removing, II and III are for water conserving.
- D. III and IV are for water removing, I and II are for conserving water.

45. What is the trophic level? It is the

- A. complex interaction between organisms in a food web.
- B. interaction of organisms with the abiotic components.
- C. linear feeding relationship between organisms.
- D. specific position an organism occupies in a food chain.

46. A grade 12 teacher demonstrated how light travelled through the eye to form an image on the retina using an eye model. Which one of the following results shows the correct sequence of this process?

- A. Reflected light → pupil → lens → optic nerve → retina → auditory nerve → brain
- B. Reflected light → sclera → choroid → lens → retina → optic nerve → brain
- C. Reflected light → cornea → pupil → lens → retina → optic nerve → brain
- D. Reflected light → cornea → pupil → lens → retina → olfactory nerve → brain

47. What is the main function of the cell membrane? It controls

- A. transport of molecules.
- B. reproduction of cells.
- C. synthesis of lipids.
- D. protein synthesis.

48. A scientist isolated three different protein molecules (X, Y and Z) from a medicinal plant as follows.

X molecule - is a segment of a protein folded into alpha-helix and beta-pleated sheet.

Y molecule - is an enzyme with its protein folded into three-dimensional structure.

Z molecule - is a linear protein with specific amino acid sequence.

Which molecule contains primary, secondary and tertiary structures of protein, respectively?

- A. Y, Z and X
- B. Y, X and Z
- C. X, Z and Y
- D. Z, X and Y

49. A student, following appropriate laboratory procedures, testing for the presence of starch in green plant leaves. He immersed the leaf in hot water for 30 seconds in one of the steps just before adding iodine. Why did he immerse the leaf into the hot water? To

- A. remove the debris and facilitate gas exchange
- B. decolorize the leaf and make it easy for the test

C. denature the enzyme and make the leaf permeable

D. improve the visibility of various pigment colors

50. Which one of the following statements explains the effects of drinking too much alcohol? It

A. exposes to constipation and esophageal disease.

B. makes hyperactive and constricts the blood vessels in the heart.

C. enables to make the right decision without fear.

D. exposes to convict domestic violence incidents.

51. This item is based on the following table that shows the feedback control mechanism from days 12 - 14 of the menstrual cycle.

No.	Hormone level	Events	Feedback system
I.	High FSH	Reduces gonadotrophin	Negative
II.	Low FSH	Menstruation	Negative
III.	High LH	Ovulation	Negative
IV.	High LH	More estradiol	Positive

Based on the above table, the correct feedback system with the events and the hormone that involved is indicated by the numbers

A. III and IV.

B. II and III.

C. I and IV.

D. I and II.

52. This item is based on the following table that shows reproduction in vertebrate animals.

No.	Birds	Frog	Rat
1.	Fertilization takes place in cloaca	Fertilization happens in the ovary	Fertilization takes place in the oviduct
2.	Magnum produces unfertilized egg	Oviduct produces fertilized egg	Ovary produces unfertilized egg
3.	Do not go through larval stage after fertilization	Go through larval stage after fertilization	Do not go through larval stage after fertilization
4.	Produce amniotic egg	Produce non-amniotic egg	Produce amniotic egg

Which alternatives contain the number that corresponds with the correct reproduction mechanism of the given vertebrates?

- A. 1, 3 and 4
- B. 1, 2 and 3
- C. 2 and 3
- D. 3 and 4

53. A survey in rural areas of Ethiopia discovered several people with enlarged body parts especially the limbs. Which one of the following is correct about the disease? It

- A. happens due to a genetic disorder that causes accumulation of fat in the lymphatic system.
- B. is caused due to bone marrow cancer that leads to deposition of fluids in the lymphatic system.
- C. happens due to deficiency of vitamins that protects lymphatic systems from damage.
- D. is caused by parasitic worms that blocks the lymphatic system leading to accumulation of fluids.

54. Why do you sense it in microsecond when a tip of your toe is knocked by obstacles?

Because the information is transmitted

- A. through electrochemical signal traveling along a neuron.
- B. through neural axon when it is in resting potential phase.
- C. by hyperpolarizing the neuronal membrane potential.
- D. due to equal charge distribution across the membrane of the axon.

55. This item is based on the following list of information about route of transport of water in plants.

- I. Entry of water from stem xylem into leaf
- II. Entry of water into the root xylem
- III. Flow of water through xylem tissues in stems
- IV. Entry of water from soil into root hairs
- V. Evaporation of water from leaf surfaces into the environment

Which one of the following alternatives shows the correct route of water in plants?

- A. IV $\Rightarrow$ II $\Rightarrow$ III $\Rightarrow$ I $\Rightarrow$ V
- B. V $\Rightarrow$ III $\Rightarrow$ II $\Rightarrow$ I $\Rightarrow$ IV
- C. V $\Rightarrow$ I $\Rightarrow$ III $\Rightarrow$ II $\Rightarrow$ IV
- D. IV $\Rightarrow$ III $\Rightarrow$ II $\Rightarrow$ I $\Rightarrow$ V

56. Which one of the following statements is correct about absorption of different end products of digestion?

- A. The last organ that absorbs nutrients is the small intestine.
- B. Amino acids are absorbed into the capillary networks.
- C. The large intestine the place for most absorption to take place.
- D. Diffusion is the only mechanism of absorbing nutrients.

57. Which of the following statements is correct about the Kyoto Protocol on climate change? It was an agreement that industrialized nations

- A. forced developing countries to restrict their carbon emissions.
- B. were obliged to reduce greenhouse gas emission significantly.
- C. encouraged to combat global epidemics by natural products.
- D. promoted urbanization in the third world countries.

58. Concerning the interactions between solar radiation and atmospheric gases, why does the accumulation of CO₂ by the fossil fuels affect the energy after reaching the Earth, rather than before? Because

- A. CO₂ is transparent to short-wave radiation but it absorbs long infrared radiation that reflects back from the surface.
- B. light energy is converted into chemical energy upon striking the soil, and then rises and creates thermal heat.
- C. fossil fuel that emits CO₂ is highly reduced comparing to the natural process that restores the balance.
- D. CO₂ acts as a catalyst that makes the sunlight 10% stronger in the upper atmosphere before getting the ground.

59. This item is based on the following information about the process of mitosis.

- I. Nuclear membrane reappears
- II. Chromosomes line up across the equatorial plane
- III. Sister chromatids pull apart by mitotic spindle
- IV. Chromosomes become thick and visible

Which one of the following alternatives contains the correct order of the above mitosis events?

- A. IV → II → III → I
- B. I → II → IV → III
- C. I → IV → III → II
- D. IV → III → II → I

60. This item is based on the following information about urine formation.

- I. Reabsorption of small solutes at the proximal convoluted tubule.
- II. Selective reabsorption of ions at the distal convoluted tubule.
- III. Filtration of the blood at the glomerulus.
- IV. Reabsorption of solutes and water at the collecting duct.
- V. Reabsorption at the loop of Henle.

Which alternative contains the correct route of urine formation in the nephrons?

- A. III → II → IV → V → I
- B. III → I → V → II → IV
- C. III → V → I → II → IV
- D. III → II → I → IV → V

61. Which of the following genetically engineered crops is herbicide-resistant? A modified crop with

- A. a foreign gene that passes to non-target crop species.
- B. an enzyme that digest plant roots of weeds.
- C. an enzyme to break down the glyphosate.
- D. an inserted gene that synthesize glyphosate in crops.

62. Which one of the following statements is correct about the pyramid of energy? It is a diagram that shows

- A. the flow energy from producers to consumers and vice versa.
- B. the flow of energy from the producers to the consumers.
- C. uniform energy transfer within all trophic level.
- D. more energy requirement for organisms at the top of the trophic level.

63. What is the effect of the deficiency of iodine?

- A. Hepatic cirrhosis and brain damage
- B. Hemosiderosis and hemochromatosis
- C. Enlargement of thyroid glands
- D. Elevation of blood pressure

64. Which alternative contains the correct comparison between the components in atmospheric air and exhaled air? Atmospheric air contains

- A. 4% carbon dioxide, but exhaled air contains 0.04% carbon dioxide.
- B. 21% oxygen, but exhaled air contains 16% oxygen.
- C. 78% nitrogen, but exhaled air contains 68% nitrogen.
- D. 15% moisture, but exhaled air contains 21% moisture.

65. Suppose a sudden earthquake starts shaking the earth while you are on a way towards your school. Which safety precaution measures should you practice?

- A. Moving towards large buildings
- B. Continue walking towards the school
- C. Standing in open ground until it stops
- D. Running back towards home

66. This item is based on the following table that shows homeotherms and poikilotherms with their respective examples.

No.	Features	Examples
1.	Have internal control over their body temperature	Frogs
2.	Use burrows to keep their body warm	Snakes
3.	Use vasoconstriction to prevent heat loss	Crocodiles
4.	Generate internal heat to adjust their body temperature	Human

Which alternative contains the numbers with the correct distinguishing features of poikilotherms and homoiotherms with their corresponding examples, respectively?

- A. 3 and 4
- B. 2 and 4
- C. 2 and 3
- D. 1 and 2

67. Viruses start replication and cause disease when they enter into their appropriate host. This is because they

- A. lack structures to generate energy and synthesize molecules.
- B. are tiny so that they lack enough space for replication.
- C. are instructed by the genetic material of living organism.
- D. lack genetic materials that instruct their vital activities.

68. Student X and Y were discussing on the origin of life on earth. Student X explained the process on how rotting meat turned into flies through time. But student Y didn't accept this view since he believed that life is the inherent property of the universe that has always existed. Based on their discussion, which views of evolution was described by the students?

- A. Panspermia by student Y and big bang theory by student X
- B. Biochemical view by student X and cosmozoic theory by student Y
- C. Special creationism by student Y and abiogenesis by student X
- D. Spontaneous generation by student X and eternity of life by student Y

69. During emergencies like car accident, individuals lose significant amount of blood that could potentially harm their health and even cost their life. What should be considered to donate blood in this condition?

- A. The donor's blood should contain antibody that protects the recipient from pathogens.
- B. The age of the donor should be under 18 years in order to donate functional bloods.
- C. The donor's and recipient's blood groups should be compatible and safe.
- D. The recipient should be young in order to receive blood from the donor.

70. Which of the following combinations, the factor and its effect on enzyme activity is correctly paired?

- A. pH - decrease kinetic energy of reacting molecules
- B. Competitive inhibitor - binds to allosteric site of an enzyme
- C. Temperature - alter the charge of amino acids
- D. Radiation - creates disorders in macromolecules

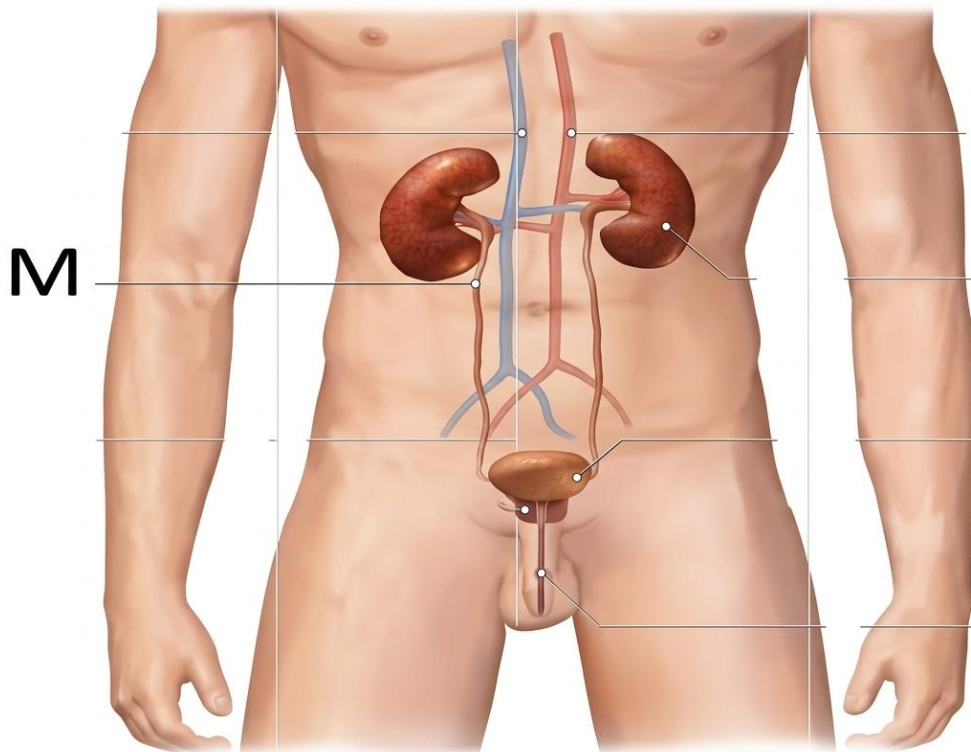
71. With regard to the contribution of soil microorganisms for soil fertility, which one of the following processes decrease nitrogen from the soil?

- A. Conversion of nitrates in the soil into nitrogen gas
- B. Fixation of atmospheric nitrogen into ammonia
- C. Decomposition of organic nitrogen sources to nitrate
- D. Transformation of ammonium ions into nitrate

72. How does gene flow become a source of genetic variations?

- A. Mass emigration of individuals stabilizes gene flow between source and receiving populations.
- B. Emigration of individuals increases genetic heterogeneity of a source population.
- C. Immigrants with new trait decreases allele frequencies of a receiving population.
- D. Immigrants with new trait increases genetic variability of a receiving population.

73. Consider the following diagram which indicates the parts of the human excretory system. Identify the part labeled 'M'.



- A. Ureter
- B. Right Kidney
- C. Urinary Bladder
- D. Urethra

74. Which one of the following is a locally available plant used for treatment of tapeworm?

- A. Cucurbita pepo (Dubba)
- B. Lepidium sativum (Feto)
- C. Eucalyptus globulus (Nech-Bahir zaf)
- D. Trigonella foenumgraecum (Abish)

75. Which one of the following shows a population pyramid of age in third world countries? A pyramid with

- A. broad base and narrowing from younger to older ages.
- B. flattened at the middle age and wider at older ages.
- C. uniform shape throughout all age groups.
- D. narrower base with younger ages.

76. Recently married woman, who wants to stay for few years without having a child, consulted a gynecologist to recommend her a contraceptive with no side effects and must be physically efficient to prevent the entry of sperm to uterus. Based on her preferences, he recommends

- A. spermicides which kill sperm and prevent the sperm from meeting the egg.
- B. oral contraceptive act by avoiding contact between sperm and ova.
- C. condom which forms barrier and works without changing the reproductive physiology.
- D. tubal ligation which prevents the contact of egg with the sperm.

77. Which of the following is correct about the cell cycle?

- A. G1 phase is the stage at which cells are metabolically inactive.

- B. M phase is the stage at which nuclear division takes place
C. S phase is the stage at which cytoplasmic division takes place.
D. G2 phase is the stage at which chromosome duplicate.

78. During photosynthesis, oxygen is produced as a by-product in

- A. the thylakoid through reduction of NADPH.
B. Calvin cycle by fixation of CO₂.
C. photosystem II by reduction of water.
D. light-dependent reaction by photolysis.

79. This item is based on the following table that shows the structure of plants and their corresponding functions.

No.	Plant structures	Functions
1.	Epidermis	Reduces water loss through transpiration
2.	Endodermis	Produces food by photosynthesis
3.	Vascular bundles	Conduct nutrients to different part of the plant
4.	Cortex	Prevent tissues from infection by pathogens

Which alternative contains the numbers with the correct plants structure and its function?

- A. 1 and 3
B. 2 and 3
C. 1 and 4
D. 2 and 4

80. What is the importance of monosaccharides? They

- A. mainly used as structural component of cells.
B. serve as the precursor of vitamin D.
C. used as stored form of energy.
D. serves as energy source for the cell.

81. Which one of the following behavioral patterns represent territoriality?

- A. Marking the directions to food sources using pheromones
B. Attracting suitable mates through visual displays
C. Surviving in an ecosystem through division of labor
D. Pounding on the chest and thumping the ground with hands

82. Which of the following is correct about pyruvate oxidation in the cell? The process involves

- A. utilizing oxygen as electron acceptor during hydroxyethyl group oxidation to form acetyl CoA.
B. decarboxylation, oxidization of hydroxyethyl group, and formation of acetyl CoA.
C. carboxylation, reduction of hydroxyethyl group and formation of acetyl CoA.
D. oxidization of hydroxyethyl group to form acetyl CoA by utilizing oxygen.

83. Which of the following is correct about bioremediation? It is the process of

- A. detoxification and degradation of pollutants using living organisms.

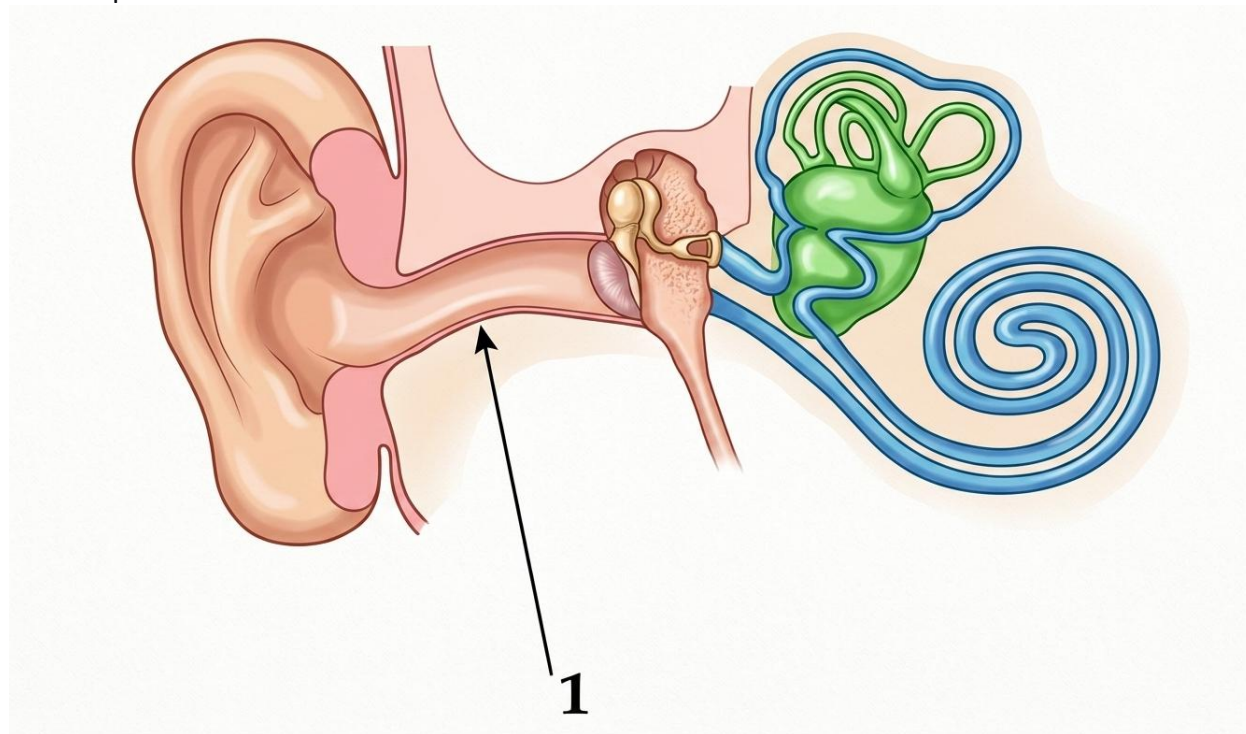
- B. composting using aerobic microorganisms to convert organic waste.
C. toxic pollutant accumulation in tissues of organisms along food chain.
D. producing biogas from inorganic wastes by aerobic microorganisms.
- 84.** Which one of the following enzymes is responsible to catalyze the joining of two molecules by hydrolysis of ATP?
- A. Ligases
B. Transferases
C. Lyases
D. Isomerases
- 85.** This item is based on the following lists of health impacts of amoebiasis and malaria.
- I. Skin ulcers and hepatosplenomegaly
II. Leads to blindness and mental retardation
III. Bloody diarrheal fluid due to intestinal cramps
IV. Vomiting, convulsion and leads to severe anemia through time
- Based on the above list, which numbers indicate the health impact of amoebiasis and malaria, respectively?
- A. II and IV
B. I and III
C. I and III
D. III and IV
- 86.** During lifting of a heavy object from the ground to certain height, the muscle
- A. contracts due to the flow of Ca^{2+} to sarcoplasmic reticulum which binds actin and myosin.
B. relaxes due to the sliding of actin and myosin by utilizing energy from ATP.
C. contracts due to the inflow of Ca^{2+} that stimulates binding of actin and myosin.
D. relaxes due to the inflow of Ca^{2+} that unbound actin and myosin.
- 87.** Which one of the following statements is correct about the role of neurotransmitters? It provides the
- A. synaptic transmission to block its one-way process by stimulating postsynaptic cell.
B. endocrine system to produces hormones that are transported in the blood plasma to their target cells.
C. nervous system to stimulate their target cells by secreting chemicals directly on to them.
D. nervous system to transmit messages between neurons, but not from neurons to muscles.
- 88.** Among the following definitions of different ecological levels, which one describes a population?
- A. A group of individuals of the same species living in a separate habitat.
B. The major terrestrial characterized by specific vegetation types and physical components.
C. A group of interbreeding organisms of the same species living in the same area at the same time.
D. The abiotic component of an area without which living things cannot survive.
- 89.** This item is based on the following description of joints with their role in movement.
- I. A joint that allows movement in all directions
II. A joint that allows movement without rotation
III. A joint that allows rotary movement when one bone swivels in a ring
IV. A joint that allows limited movement by slipping

V. A joint that allows movement in one direction along one plane only

Based on the above information, condyloid joint, pivot joint and hinge joint are respectively represented by numbers

- A. I, II and V.
- B. II, III and V.
- C. II, III and IV.
- D. I, III and V.

90. Consider the following diagram, which indicates the parts of human ear. Identify the labeled part '1'.



- A. Tympanic membrane (Eardrum)
- B. Auditory canal
- C. Cochlea
- D. Semicircular canals

91. A hemophilic man has two daughters and two sons from neither carrier nor affected woman. One of their sons married to a girl who is heterozygous to hemophilia. What is the probability that one of their daughters become hemophilic?

- A. 50%
- B. 0%
- C. 75%
- D. 25%

92. The scientific names of which one of the following plants is properly written?

- A. Tef - Eragrostis Tef
- B. Maize - Zea mays
- C. Wheat - triticum aestivum
- D. Enset - ensete Ventricosum

93. Which of the following blood vessels carry oxygenated and deoxygenated into and away from the kidneys?

- A. Iliac artery and iliac vein
- B. Hepatic artery and hepatic vein
- C. Renal artery and renal vein
- D. Pulmonary artery and pulmonary vein

94. Grade 10 student placed seeds 'X' and 'Y' in separate containers containing soil and other requirements of germination. From her observation, the cotyledon of seed 'X' remained under the soil during germination, whereas that of seed 'Y' was pushed above the soil surface. What would she conclude about the experiment?

- A. Seed 'X' was most probably monocot whereas seed 'Y' was a dicot.
- B. Seed 'Y' has hypogea, whereas seed 'X' has epigeal germination.
- C. Differences in the germination was related with availability of water.
- D. Light intensity was responsible to grow above and below the soil.

95. Which of the following is correct about founder effect in a population? It occurs when

- A. the size of a population is severely reduced due to a sudden accident.
- B. new group fully represents genetic diversity of the original population.
- C. the new small colony is established from the original population.
- D. a natural disaster kills most individuals of the original population.

96. Which one of the following indigenous practices in Ethiopia is correctly matched with the purpose of plants conservation?

- A. Planting in graveyards - for preserving crop varieties
- B. Crop rotation - for preserving indigenous woody plants
- C. Protection in sacred places - preserving genetic resources of vegetables
- D. Agroforestry - maintaining the genetic diversity of crops and trees

97. Which of the following parts of axial skeleton is correctly matched with its function?

- A. Hyoid - protects heart and the lungs
- B. Cranium - protects the brain
- C. Sternum - amplify sound wave
- D. Ribs - holds up human body

98. Suppose you have given a wing of bird, a wing of insect and a forelimb of a mammal to study their ancestral relationship, then which of the following statements is correct regarding their evolutionary relationship?

- A. These structures have functional similarities that evolved from the same ancestor.
- B. Forelimbs of mammals and wings of insects have structural similarities confirming that they are closely related.
- C. Wings of birds and forelimbs of mammals are homologous structures that were evolved from similar origin.
- D. Wings of bird and insects are analogous structures that were evolved from similar origin.

99. Which property of an enzyme and its description is correctly matched?

- A. Specificity - the ability of an enzyme to catalyze different substrates at a time
- B. Regulation - controlling activity of an enzyme by activators or inhibitors
- C. Reversibility - the ability of an enzyme to repair itself after denaturation
- D. Biocatalytic property - the ability of an enzyme to dissolve in solvents

100. Which of the following best describes the outcome of meiosis in human cells?

- A. Two diploid daughter cells identical to the parent cell
- B. Four haploid daughter cells, each with half the chromosome number of the parent cell
- C. Two haploid cells that immediately fuse to form a zygote
- D. Four diploid cells that undergo further division

Answer key

1. Answer: A

Explanation: While using bioengineered bacteria has environmental risks, adopting the technology with **stringent safety protocols** is the scientifically valid approach to mitigate those risks while addressing the plastic waste problem.

2. Answer: C

Explanation: The correct path of hearing is: **Pinna collects sound (III) → Eardrum vibrates (IV) → Ossicles amplify vibrations (II) → Nerve impulses sent to brain (I)**. The correct sequence is III → IV → II → I.

3. Answer: B

Explanation: **Sericulture** (the rearing of silkworms) is a major industrial process used to produce silk fibers for the textile industry.

4. Answer: D

Explanation: Process A (ribosome reading mRNA to make a protein) is **Translation**. Process B (mRNA synthesis in the nucleus) is **Transcription**.

5. Answer: B

Explanation: Covalent modification regulates enzyme activity through the **addition or removal** of specific chemical groups (e.g., phosphate, methyl) to/from the enzyme protein.

6. Answer: C

Explanation:

- ✓ **1 (Pepsin)** - Hydrolyzes proteins to peptides (often simplified to amino acids in exams), pH 1.
- ✓ **2 (Amylase)** - Correctly converts starch to maltose at pH 4-6.
- ✓ **4 (Lactase)** - Correctly converts lactose to glucose and galactose at pH 7-8.
- ✓ **3 (Nuclease)** is incorrect because nuclease hydrolyzes nucleic acids into nucleotides, not the reverse. Thus, 1, 2, and 4 are correct.

7. Answer: B

Explanation: A cross between a heterozygote (**Rr**) and a homozygous recessive (**rr**) yields **50% Rr** (resistant) and **50% rr** (non-resistant).

8. Answer: D

Explanation: Resolving conflict caused by overpopulation is a **sociological/political issue**, not a direct application of biological discoveries or tools.

9. Answer: C

Explanation: Ovaries secrete **estrogen**, which promotes the maturation of the ovum (egg), while testes produce **testosterone** and sperm.

10. (Omitted – Promotional message in original file)

11. Answer: C

Explanation: A hypertonic solution causes water to leave the plant cell. The vacuole shrinks, and the cytoplasm pulls away from the cell wall—a process called **plasmolysis**.

12. Answer: B

Explanation: Photorespiration occurs when Rubisco fixes O₂ instead of CO₂. This leads to the **oxidation of organic compounds** and the release of CO₂ without producing ATP.

13. Answer: A

Explanation: Honey bees staying in their hive to conserve heat is a **behavioral** adaptation. The other options describe physiological (involuntary) responses.

14. Answer: A

Explanation: Hillside terracing is a **mechanical** conservation method because it physically intercepts and slows down runoff water draining from sloped areas, preventing soil erosion.

15. Answer: D

Explanation: **Specific heat capacity** is defined as the energy required to raise the temperature of 1 gram of a substance by 1°C.

16. Answer: D

Explanation: **1 (Sucrose)** is correctly composed of glucose + fructose. **4 (Lipids)** are correctly composed of fatty acids + glycerol. *Starch (2) is made only of glucose, and Proteins (3) contain nitrogen (not just C, H, O).*

17. Answer: C

Explanation: Bioethics deals with the moral implications of applying scientific discoveries (e.g., cloning, gene editing, stem cells) to **human wellbeing**.

18. Answer: A

Explanation: The classic symptoms of **gonorrhea** include **painful urination** and a purulent (abnormal) discharge from the genitalia.

19. Answer: D

Explanation: **Metabolism** is the sum total of all chemical reactions in the body, including both catabolism (breakdown) and anabolism (synthesis).

20. Answer: C

Explanation: The **Aorta** (labeled 4 in the diagram) receives oxygenated blood from the left ventricle and distributes it to the entire body. *(A is wrong: left atrium receives oxygenated blood; B is wrong: right ventricle pumps deoxygenated blood; D is wrong: pulmonary vein carries oxygenated blood).*

21. Answer: B

Explanation: Sister chromatids lining up end-to-end at the equator = **Metaphase II**. Chromatids being pulled to opposite poles = **Anaphase II**.

22. Answer: D

Explanation: **Vitamin C** is essential for synthesizing collagen. Deficiency leads to **scurvy**, causing bleeding gums and poor wound healing.

23. Answer: C

Explanation: Unilateral light causes **phototropism**. Auxin (plant hormone) accumulates on the shaded side, promoting faster cell elongation there and causing the plant to bend toward the light source.

24. Answer: B

Explanation: The population growth slows and levels off as it reaches the environment's carrying capacity—this is the classic **logistic growth** model (S-curve).

25. Answer: A

Explanation: Organisms with a segmented body, jointed appendages (six legs), and an exoskeleton belong to the phylum **Arthropoda**.

26. Answer: D

Explanation: Innate immunity is the body's **non-specific** first line of defense. It responds the same way to all pathogens and lacks immunological memory.

27. Answer: D

Explanation: **Microbes** are not pollinating agents. Pollination is carried out by wind, insects, birds, bats, and water.

28. Answer: B

Explanation: A **mutation** (which can be induced by mutagens or occur spontaneously) is the random, sudden change in the DNA sequence that drives evolution.

29. Answer: A

Explanation: When a non-native species is introduced and outcompetes native species, it is classified as an **invasive species**, causing biodiversity loss.

30. Answer: A

Explanation: The **trachea** is reinforced with C-shaped cartilaginous rings which prevent the airway from collapsing during breathing.

31. Answer: B

Explanation: $BMI = 68 / (1.81)^2 \approx 20.7$. A BMI between 18.5 and 24.9 is classified as **normal weight**.

32. Answer: C

Explanation: In **complete metamorphosis**, eggs hatch into distinct, worm-like **larvae** (e.g., caterpillars) after a few days, which then pupate before becoming adults.

33. Answer: B

Explanation: Plant cells are distinguished from animal cells by having a **large central vacuole** for storage and turgor pressure, plus a cell wall and chloroplasts.

34. Answer: D

Explanation:

- ✓ **1 (Grass)** correctly fits as a Producer (1st level).
- ✓ **3 (Snake)** is correctly described as a Tertiary consumer (4th level) feeding on a carnivore (frog).
- ✓ **4 (Grasshopper)** is correctly described as a Primary consumer (2nd level) feeding on green plants.
- ✓ **2 (Frog)** is wrong because it feeds on herbivores, making it a Secondary consumer, not Tertiary.

35. Answer: A

Explanation: **Bacilli** correctly refers to rod-shaped bacteria. (Spirochaetes are spiral, Cocci are spherical).

36. Answer: C

Explanation: Severe deficiency of **Calcium** (and Vitamin D) in children leads to **rickets**, causing soft, weakened bones.

37. Answer: B

Explanation: The ETS produces the most ATP because it utilizes the stored potential energy

in **NADH and FADH₂** to pump protons across the membrane, creating a gradient that powers ATP synthase.

38. Answer: D

Explanation: **Gene therapy** works by introducing, removing, or altering **sections of DNA** in a patient's cells to treat or cure genetic disorders.

39. Answer: A

Explanation: **Molars** have a broad, flat surface specifically adapted for crushing and grinding food. (Canines tear, Incisors cut, Premolars crush/grind).

40. Answer: C

Explanation: Fungi are heterotrophic organisms that digest their food externally by secreting enzymes into the environment (**extracellular digestion**) and then absorbing the nutrients.

41. Answer: B

Explanation: A **mutation** is fundamentally a random, often spontaneous change in the genetic information (DNA) of an organism.

42. Answer: C

Explanation: To physically isolate individual bacteria from a liquid sample, the standard next step is to **perform serial dilutions** and streak/plate them on a growth medium to obtain isolated colonies.

43. Answer: B

Explanation: **Oxytocin** is the hormone responsible for stimulating powerful uterine contractions during childbirth.

44. Answer: D

Explanation:

- ✓ **High ADH (II)** makes the tubules permeable to water (**I**) → **Water conserving**.
- ✓ **Very low ADH (IV)** reduces reabsorption, producing dilute urine (**III**) → **Water removing**.
- ✓ Thus, III & IV are for water removing; I & II are for water conserving.

45. Answer: D

Explanation: A **trophic level** is the specific feeding position an organism occupies within a food chain (e.g., Producer, Primary Consumer, Secondary Consumer).

46. Answer: C

Explanation: The correct path of light through the eye is: **Cornea → Pupil → Lens → Retina → Optic nerve → Brain**.

47. Answer: A

Explanation: The cell membrane is selectively permeable, so its primary function is to control the **transport of molecules** into and out of the cell.

48. Answer: D

Explanation:

- ✓ **Z** (linear sequence) = Primary structure.
- ✓ **X** (alpha-helix/beta-sheet) = Secondary structure.
- ✓ **Y** (3D folded enzyme) = Tertiary structure.
- ✓ Order: Z → X → Y.

49. Answer: C

Explanation: Immersing the leaf in boiling water kills the cells, **denatures the enzymes**, and makes the cell membrane permeable so that iodine can penetrate and react with the starch.

50. Answer: D

Explanation: Alcohol reduces inhibitions and impairs judgment, which significantly increases the risk of involvement in **domestic violence incidents**.

51. Answer: A

Explanation: During days 12–14 of the menstrual cycle, **high levels of estradiol (from the developing follicle) stimulate a surge in LH and FSH**. This is a classic **positive feedback** loop. **III** (High LH leading to ovulation) and **IV** (High LH stimulating more estradiol via positive feedback) are correctly identified.

52. Answer: A

Explanation:

- ✓ **1** is correct: Rat fertilization occurs in the oviduct.
- ✓ **3** is correct: Birds and Rats do not go through a larval stage (unlike frogs).
- ✓ **4** is correct: Birds and Rats produce amniotic eggs (unlike frogs).
- ✓ **2** is incorrect because birds produce unfertilized eggs in the ovary, not the magnum.

53. Answer: D

Explanation: The condition described is **elephantiasis (lymphatic filariasis)**. It is caused by parasitic filarial worms that block the lymphatic system, leading to severe fluid accumulation and massive swelling (edema) in the limbs and other body parts.

54. Answer: A

Explanation: The nervous system transmits information at extremely high speeds via **electrochemical signals (action potentials)** traveling along neurons.

55. Answer: A

Explanation: The correct route of water in plants is: **Soil → root hairs (IV) → root xylem (II) → stem xylem (III) → leaf xylem (I) → evaporation into the environment (V)**.

56. Answer: B

Explanation: Amino acids (along with monosaccharides) are absorbed into the **capillary networks** within the villi of the small intestine. (*The small intestine is the primary site of absorption; the large intestine absorbs water/minerals*).

57. Answer: B

Explanation: The **Kyoto Protocol** was an international treaty that legally obligated **industrialized nations** to significantly reduce their greenhouse gas emissions.

58. Answer: A

Explanation: **CO₂ is a greenhouse gas**. It is transparent to incoming short-wave solar radiation but absorbs outgoing long-wave infrared radiation (heat) re-emitted by the Earth's surface, trapping it in the atmosphere.

59. Answer: A

Explanation: The correct order of mitosis is:

- ✓ **IV**: Chromosomes become thick and visible (**Prophase**)
- ✓ **II**: Chromosomes line up at the equator (**Metaphase**)
- ✓ **III**: Sister chromatids pull apart (**Anaphase**)
- ✓ **I**: Nuclear membrane reappears (**Telophase**)

60. Answer: B

Explanation: The correct route of urine formation in the nephron is: **Glomerulus (III) →**

Proximal convoluted tubule (I) → Loop of Henle (V) → Distal convoluted tubule (II) → Collecting duct (IV).

61. Answer: C

Explanation: Herbicide-resistant crops (like "Roundup Ready") are genetically modified to contain an **enzyme that breaks down the herbicide glyphosate**, rendering the crop immune to it.

62. Answer: B

Explanation: A **pyramid of energy** illustrates the directional flow of energy from **producers at the base up to the consumers** at higher trophic levels. Energy decreases at each step (roughly 90% is lost as heat).

63. Answer: C

Explanation: Iodine is required to synthesize thyroid hormones. When iodine is deficient, the thyroid gland enlarges (hypertrophies) as it attempts to trap more iodine, resulting in a **goiter**.

64. Answer: B

Explanation: Atmospheric (inspired) air contains approximately **21% oxygen**, while exhaled air contains about **16% oxygen** (the rest is converted to CO₂ or remains unused).

65. Answer: C

Explanation: During an earthquake, if you are outdoors, the safest measure is to **stand in an open ground** away from buildings, trees, and power lines to avoid falling debris.

66. Answer: B

Explanation:

- ✓ **2** matches a **poikilotherm** (ectotherm) - Snakes use burrows to regulate their body temperature.
- ✓ **4** matches a **homeotherm** (endotherm) - Humans generate internal heat to maintain a constant body temperature.

67. Answer: A

Explanation: Viruses are obligate intracellular parasites. They **lack structures (ribosomes, mitochondria) to generate energy and synthesize molecules**, so they must hijack the host cell's machinery to replicate.

68. Answer: D

Explanation:

- ✓ **Student X** (flies from rotting meat) described the **Spontaneous Generation** theory.
- ✓ **Student Y** (life has always existed) described the **Eternity of Life** theory.

69. Answer: C

Explanation: Blood transfusions require strict **compatibility of blood groups** (ABO and Rh systems) between the donor and recipient to prevent fatal agglutination (clotting) reactions.

70. Answer: D

Explanation: High-energy radiation (like UV or X-rays) can break chemical bonds and cause structural damage (**create disorders**) in enzyme macromolecules, leading to denaturation.

71. Answer: A

Explanation: **Denitrifying bacteria** convert nitrates in the soil back into atmospheric nitrogen gas (N₂). This process **decreases** the nitrogen available to plants in the soil.

72. Answer: D

Explanation: Gene flow occurs when immigrants with new genetic traits enter a population. This **increases the genetic variability** of the receiving population by introducing new alleles.

73. Answer: A

Explanation: In the diagram of the excretory system, the part labeled '**M**' is the **Ureter**, the tube that carries urine from the kidney to the urinary bladder.

74. Answer: A

Explanation: The seeds of **Cucurbita pepo** (pumpkin/Dubba) are a locally available traditional remedy used to treat tapeworm infections.

75. Answer: A

Explanation: Third-world (developing) countries typically have an **expansive population pyramid** characterized by a **broad base** (high birth rates) that narrows rapidly toward the older age groups (higher death rates).

76. Answer: C

Explanation: A **condom** is a physical barrier that prevents sperm from entering the uterus. It has **no side effects** on reproductive physiology and works purely mechanically.

77. Answer: B

Explanation: The **M phase** (Mitosis) is the stage of the cell cycle where **nuclear division** occurs, separating the duplicated chromosomes.

78. Answer: D

Explanation: During the **light-dependent reactions**, water molecules are split (**photolysis**) to provide electrons and protons. **Oxygen (O₂) is released as a by-product** of this process.

79. Answer: A

Explanation:

- ✓ **1 (Epidermis):** Correctly reduces water loss through transpiration.
- ✓ **3 (Vascular bundles):** Correctly conduct nutrients and water to different parts of the plant.
- ✓ *(2 and 4 are incorrect functions for endodermis and cortex).*

80. Answer: D

Explanation: Monosaccharides (like glucose) are the simplest form of carbohydrates and serve as the **primary, immediate energy source** for cellular respiration.

81. Answer: D

Explanation: **Territoriality** is a behavior used to defend a specific area. **Pounding on the chest and thumping the ground** is an aggressive display used by primates to establish and defend their territory.

82. Answer: B

Explanation: Pyruvate oxidation (the link reaction) involves:

1. **Decarboxylation** (removal of a CO₂).
2. **Oxidization** of the hydroxyethyl group.
3. **Formation of Acetyl CoA** by attaching the remaining 2-carbon molecule to Coenzyme A.

83. Answer: A

Explanation: **Bioremediation** is the use of living organisms (such as bacteria, fungi, or plants) to **detoxify and degrade** environmental pollutants.

84. Answer: A

Explanation: **Ligases** are enzymes that catalyze the joining (ligation) of two large molecules by forming a new chemical bond, a process usually powered by the hydrolysis of ATP.

85. Answer: D

Explanation:

- ✓ **Amoebiasis** causes severe intestinal inflammation leading to **bloody diarrheal fluid** and cramps (III).
- ✓ **Malaria** causes symptoms like vomiting, convulsions, and can lead to **severe anemia** over time (IV).

86. Answer: C

Explanation: Muscle contraction is triggered by the **inflow of Ca^{2+}** ions into the sarcoplasm. Calcium binds to troponin, which exposes the binding sites on actin, allowing **actin and myosin to bind** and slide past each other.

87. Answer: C

Explanation: Neurotransmitters are chemical messengers secreted by neurons directly onto target cells (neurons, muscles, or glands) to stimulate or inhibit them, facilitating **nervous system communication**.

88. Answer: C

Explanation: A **population** in ecology is defined as a group of **interbreeding individuals of the same species** living in the same geographic area at the same time.

89. Answer: B

Explanation:

- ✓ **Condylloid joint:** Allows movement without rotation (II).
- ✓ **Pivot joint:** Allows rotary movement when one bone swivels in a ring (III).
- ✓ **Hinge joint:** Allows movement in one direction along one plane only (V).

90. Answer: B

Explanation: In the diagram of the ear, the part labeled '1' is the **Auditory canal** (or ear canal), which funnels sound waves from the pinna to the tympanic membrane (eardrum).

91. Answer: B

Explanation: The hemophilic man (X^hY) marries a normal woman (XX). Their sons inherit the Y from the father and an X from the mother, so all sons are **normal (XY)**. One normal son (XY) marries a heterozygous carrier girl (X^hX). Their daughters will inherit the father's normal X (X) and either the mother's normal X or carrier X, so they will be **either normal (XX) or carriers (X^hX), but never hemophilic (X^hX^h)**. Probability is **0%**.

92. Answer: B

Explanation: In binomial nomenclature, the **genus name is capitalized** and the **species name is lowercase** (e.g., *Zea mays*). The others have capitalization errors or improper formatting.

93. Answer: C

Explanation: The **renal artery** carries oxygenated blood *into* the kidneys, and the **renal vein** carries deoxygenated (filtered) blood *away* from the kidneys.

94. Answer: A

Explanation:

- ✓ **Seed X** (cotyledon remains under soil, hypogeal) is typical of **monocots** (e.g., maize).
- ✓ **Seed Y** (cotyledon pushed above soil, epigeal) is typical of **dicots** (e.g., beans).

95. Answer: C

Explanation: The **founder effect** occurs when a **small new colony** is established by a few individuals from a larger original population, leading to a loss of genetic diversity in the new population.

96. Answer: D

Explanation: **Agroforestry** is an indigenous practice that combines trees and crops. It effectively maintains the **genetic diversity** of both crop plants and trees by creating sustainable, diverse land-use systems.

97. Answer: B

Explanation: The **Cranium** (skull) is a part of the axial skeleton, and its main function is to **protect the brain**.

98. Answer: D

Explanation: The wings of a bird and the wings of an insect have the same function (flight) but evolved from completely different ancestral origins. They are **analogous structures**, not homologous.

99. Answer: B

Explanation: **Regulation** is the property of an enzyme that allows its activity to be controlled by **activators** (which speed it up) or **inhibitors** (which slow it down).

100. Answer: B

Explanation: **Meiosis** is a reduction division process. It results in **four haploid daughter cells**, each containing exactly **half the chromosome number** (n) of the original diploid parent cell ($2n$).

